5 **(a)** State what is meant by the mass defect of a nucleus.

Mass defect is defined as the difference between the total mass of the separate nucleons in a nucleus [B1] and the combined mass of the nucleus [B1]

[2]

(b) A stationary nucleus of uranium-238 ($^{238}_{92}U$) undergoes alpha decay to produce an excited nucleus of thorium-234 ($^{234}_{90}Th^{*}$). The kinetic energy of the emitted alpha particle is 4.151 MeV. A gamma-ray photon is then immediately emitted from the excited thorium nucleus after the alpha decay.

Assume that the rebound kinetic energy of the thorium nucleus is negligible.

Fig. 5.1 shows the masses of some of the nuclides involved in the decay reaction.

nuclide	nuclide mass/ u
$^{4}_{2}\alpha$	4.000407
$^{234}_{90}Th$	233.915174

Fig. 5.1

The total energy released in the decay of the nucleus of uranium-238 is 4.274 MeV.

(i) Calculate the mass, in u, of the uranium-238 nuclide.

$$E = \Delta mc^2$$

$$4.274 \times 10^6 \times 1.60 \times 10^{-19} = \Delta m(3.00 \times 10^8)^2 \quad \text{[C1]}$$

$$\Delta m = 7.598 \times 10^{-30} \text{ kg}$$

$$= \frac{7.598 \times 10^{-30}}{1.66 \times 10^{-27}} \text{ u}$$

$$= 0.004577 \text{ u}$$

$$m_U = 0.004577 + 233.915174 + 4.000407 \quad \text{[C1]}$$

$$= 237.920158 \text{ u} \quad \text{(6 d.p.)[A1]}$$

mass = ... u [3]

(ii) Determine a value for the wavelength of the gamma radiation emitted during the decay of the uranium-238 nucleus.

$$\text{Total Energy released} = KE_{Th} + KE_{\alpha} + E_{\gamma}$$

$$4.274 \times 10^{6} \times 1.60 \times 10^{-19} = 0 + 4.151 \times 10^{6} \times 1.60 \times 10^{-19} + E_{\gamma} \quad \text{[C1]}$$

$$E_{\gamma} = 1.968 \times 10^{-14} = \frac{hc}{\lambda} \quad \text{[C1]}$$

$$1.968 \times 10^{-14} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{\lambda}$$

$$\lambda = 1.011 \times 10^{-11} \text{ m} \quad \text{(4.s.f)} \quad \text{[A1]}$$

wavelength = ... m [3]

(iii) In practice, the rebound kinetic energy of the thorium nucleus is **not** negligible.

Explain, without further calculation, how your answer in **(b)(ii)** compares with the true wavelength of gamma radiation emitted during the decay of the uranium-238 nucleus.

With Thorium having some kinetic energy, the true energy of the gamma photon would be smaller, so the true wavelength would be larger in practice ($E = \frac{hc}{\lambda}$). [B1]

[1]

(c) Gamma radiation emitted during the decay of a sample of uranium-238 has a single wavelength.

Nuclei of cobalt-60 ($^{60}_{27}Co$) decay by beta emission and also emit gamma radiation in the process.

Suggest why there is **not** a single wavelength for the gamma radiation emitted during the decay of a sample of cobalt-60.

In cobalt-60 beta decay, (anti)neutrinos are emitted. [B1]

Total energy released is shared between the daughter nucleus, beta particle and (anti)neutrino. Since there are many ways that the energy can be divided among the three particles to satisfy the conservation principles, multiple possible energies for daughter nucleus will allow for multiple possible energies for gamma radiation and thus variable wavelengths. [B1] [2]

[Total: 11]